LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (20)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	20	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to electronic instruments and advantages
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to electronic instruments and advantages.
5. Specific Objectives	20.3.1.1 Trainee will be able to explain the construction and operation of DMM & CRO.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of DMM and CRO front panel controls; Safety: High voltage caution; Supervision: Instructor monitors connections.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an electronic instrument?
(b) Name two advantages of electronic instruments over mechanical ones.
(c) Why is calibration important for measurement accuracy?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of electronic instruments • Advantages over mechanical types • General construction principles • Signal conditioning basics • Accuracy and precision concepts	L & Ds	PPT, IB	15 min
(b)	• Digital Multimeter (DMM) construction • DMM operation modes • Cathode Ray Oscilloscope (CRO) construction • CRO operation principles • Comparison of DMM and CRO	L & Ds	PPT, PDF	20 min
(c)	• Safety precautions for electronic instruments • Common errors in measurement • Maintenance and storage requirements • Troubleshooting basic faults • Summary of key advantages	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the main advantage of a DMM over an analog multimeter.
(b) Describe the role of the electron beam in a CRO.
(c) List three safety precautions when operating high-voltage instruments.

12. Home Assignment and Follow Up	Prepare a brief note on how the knowledge of DMM and CRO principles will assist in the study of Spectrum analyzers and function generators in the next lesson plan.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Functions and operation of DMM, Spectrum analyzer and function generator.	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min